Rare process biasing study & production plan

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Why study biasing factor

- We use biasing to enhance the production rate of important process
- To obtain the actual rare process rate:
  - From inclusive sample: large stat. unc.
  - Theoretical calculation: non-trivial
  - Biasing factor Fit ✅

Probability of rare process

Using $dn_1/dt = -\langle\sigma v\rangle n_1 n_2$. For a ensemble of biasing particles with density n_1 , and same velocity: $dn_1/vdt = -\langle\sigma\rangle n_1 n_2$

Distribution of interaction length L_i:

$$P_{\langle\sigma_i\rangle}\left(L_i\right) = \langle\sigma_i\rangle n e^{-\langle\sigma_i\rangle n L_i}$$

Probability of rare process happening:

$$\begin{aligned} P_{rare} &= P(L_r < L_e \,\&\, L_r < D) \\ &= P(L_e < D \,\&\, L_r < L_e) + P(L_e > D \,\&\, L_r < D) \\ &= \int_0^D dL_e P_{\langle\sigma_e\rangle}\left(L_e\right) \int_0^{L_e} dL_r P_{\langle\sigma_r\rangle}\left(L_r\right) + \int_D^{+\infty} dL_e P_{\langle\sigma_e\rangle}\left(L_e\right) \int_0^D dL_r P_{\langle\sigma_r\rangle}\left(L_r\right) = \frac{\langle\sigma_r\rangle}{\langle\sigma_r\rangle + \langle\sigma_e\rangle}\left(1 - e^{-(\langle\sigma_r\rangle + \langle\sigma_e\rangle)nD}\right) \end{aligned}$$

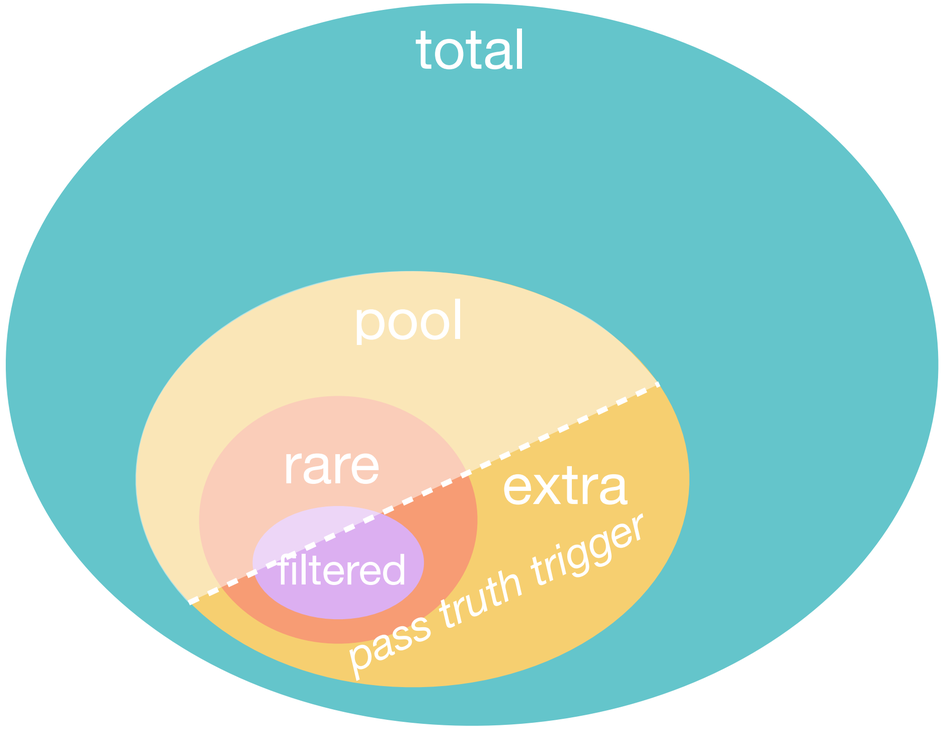
With bias factor x: $\langle\sigma_r\rangle \to x\langle\sigma_r\rangle$, and let $k \equiv \frac{\langle\sigma_r\rangle}{\langle\sigma_e\rangle}, p \equiv \langle\sigma_e\rangle nD,$

$$P(x) = \frac{kx}{kx+1}\left(1 - e^{-p(kx+1)}\right)$$

Approximation: 1-dimensional

Event classification in biasing study

- **total:** Total beam on events
- **pool:** Produced **biasing particle**, and biasing particle pass BiasEmin cut
- **rare:** Successfully generate **biasing process**.
- **extra:** pool - rare
- With Bias Factor ↑, N_{rare} ↑, until $N_{rare} = N_{pool}$ at saturation.
- Sometimes we add a final state cut (i.e. ΔE>2 GeV for EN), and only record **filtered** events.



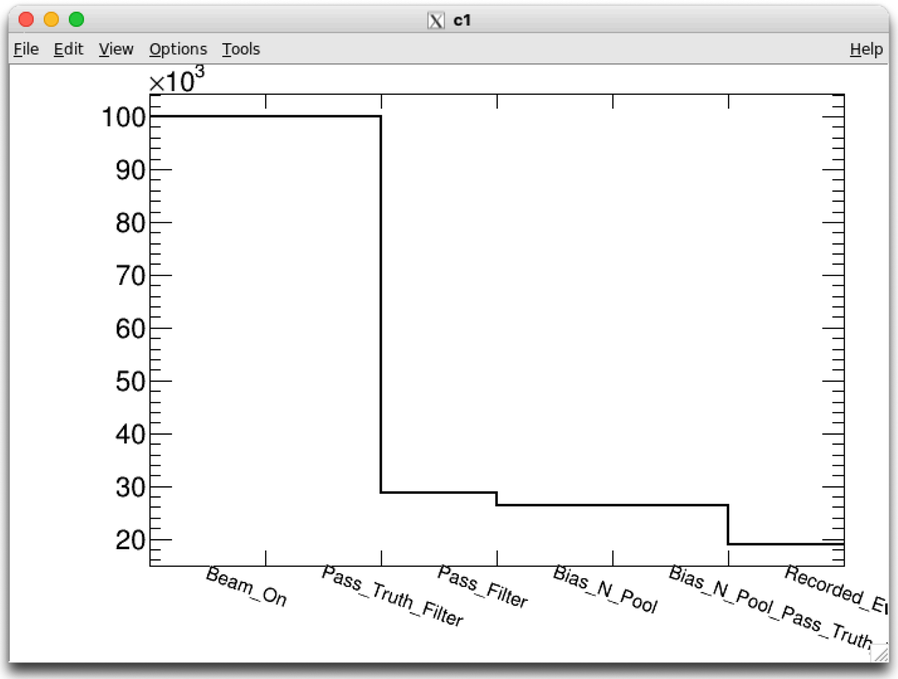
$$N_{rare} = Rate(rare) \times \epsilon_{filter} \times N_{EOT}$$

$$Rate(x) = \frac{n_{pool}}{n_{total}} \times P_{rare}(x)$$

Methodology

Git repo

- Scan bias factor: use N_{rare}/N_{pool} to fit P(x)
- Use N_{pool} for each event instead of constant n_{pool} extracted by larger sample to eliminate statistical fluctuation.
- Add histogram truth_cutflow in DSimu output, to extract key data:
  - N_{pool} , N_{rare} , filter efficiency
  - Compatible with hadd merging file
- Apply filter (Apply trigger)
- Limit to at most 1 biased interaction in each event
- Random seed different across process type

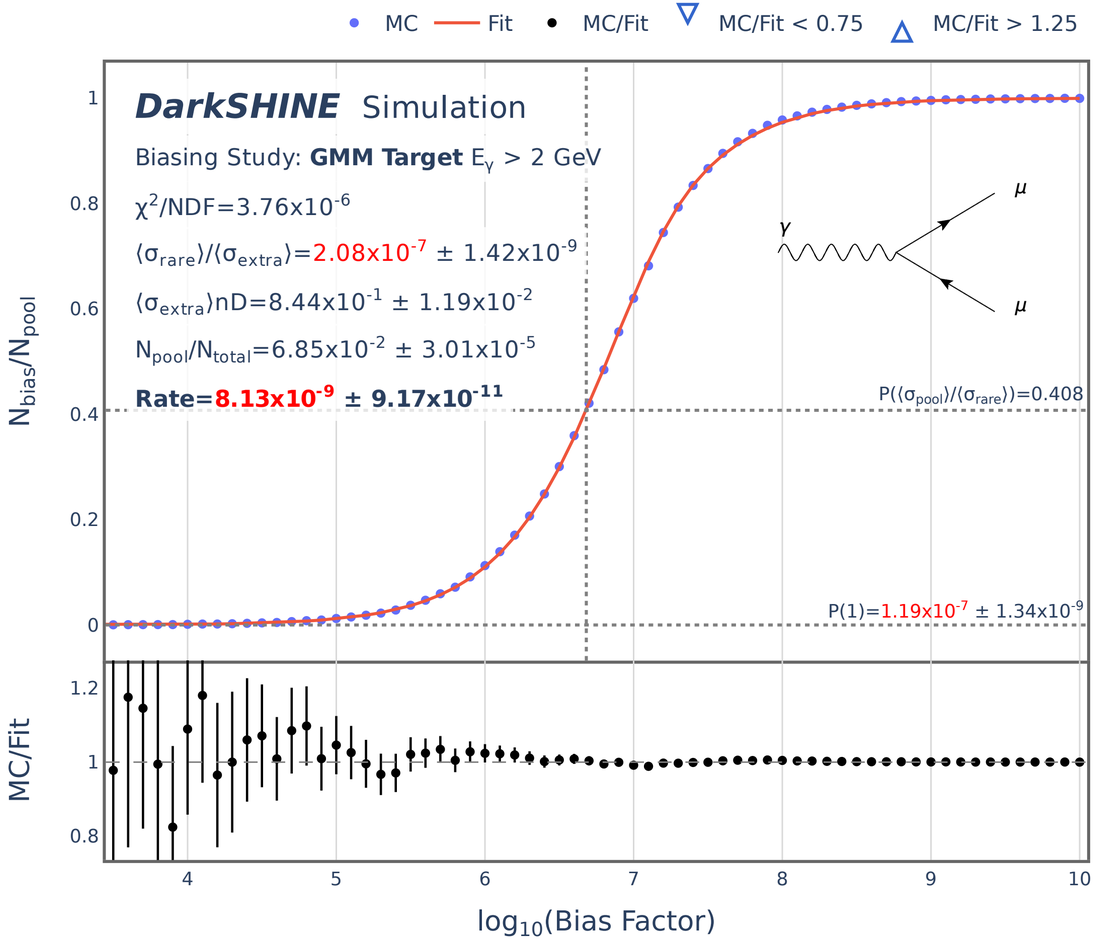


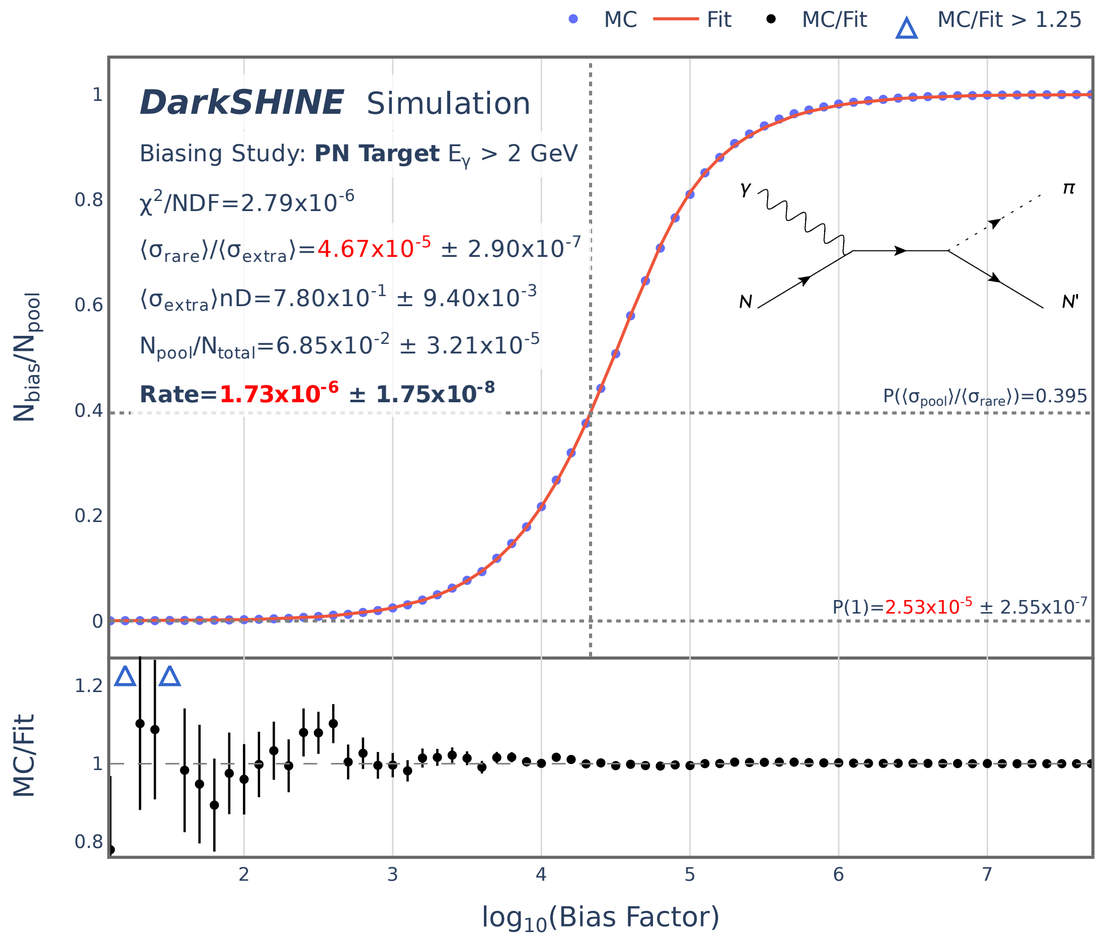
$$P(x) = Br(x) \times \left(1 - e^{-(x\langle\sigma_r\rangle + \langle\sigma_e\rangle)nD}\right)$$

Target γ process

P(1) < Br

$\langle\sigma\rangle nD$ difference: maybe angular difference in phase space?

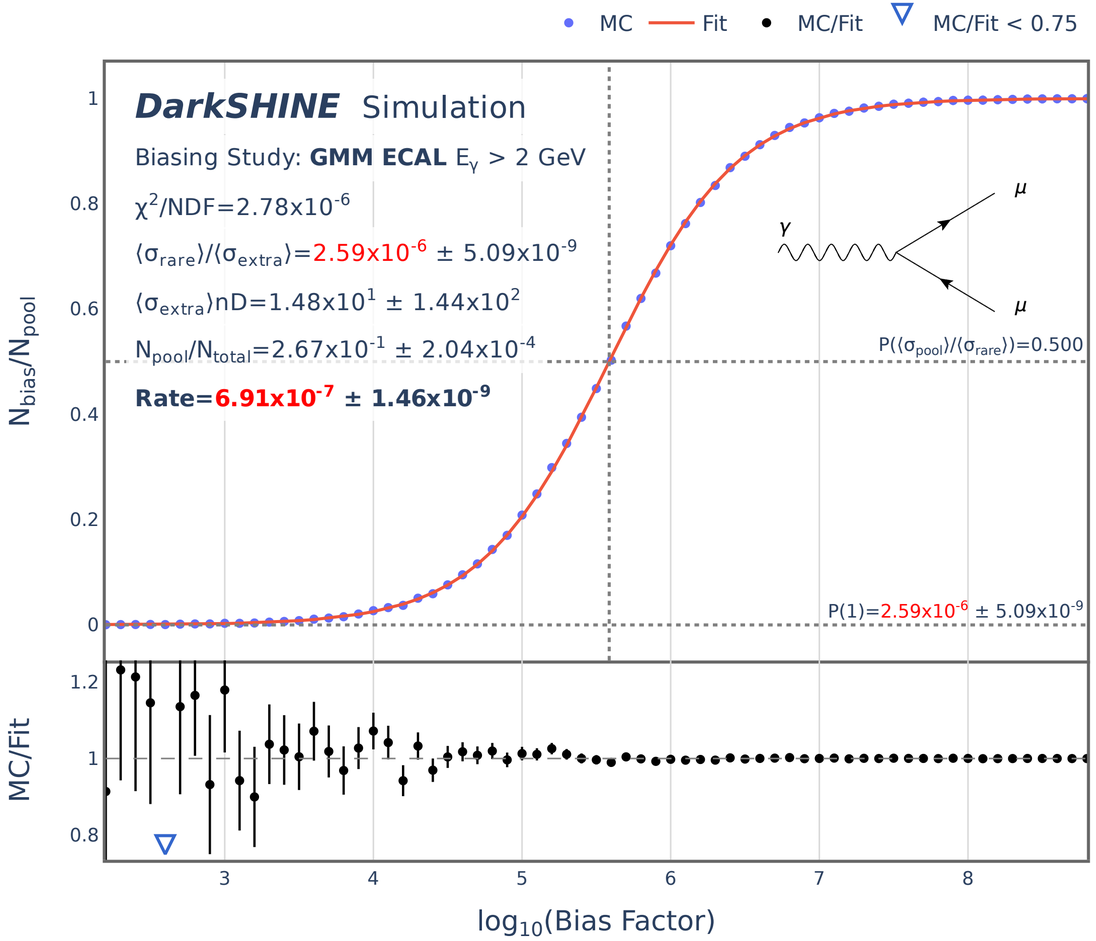


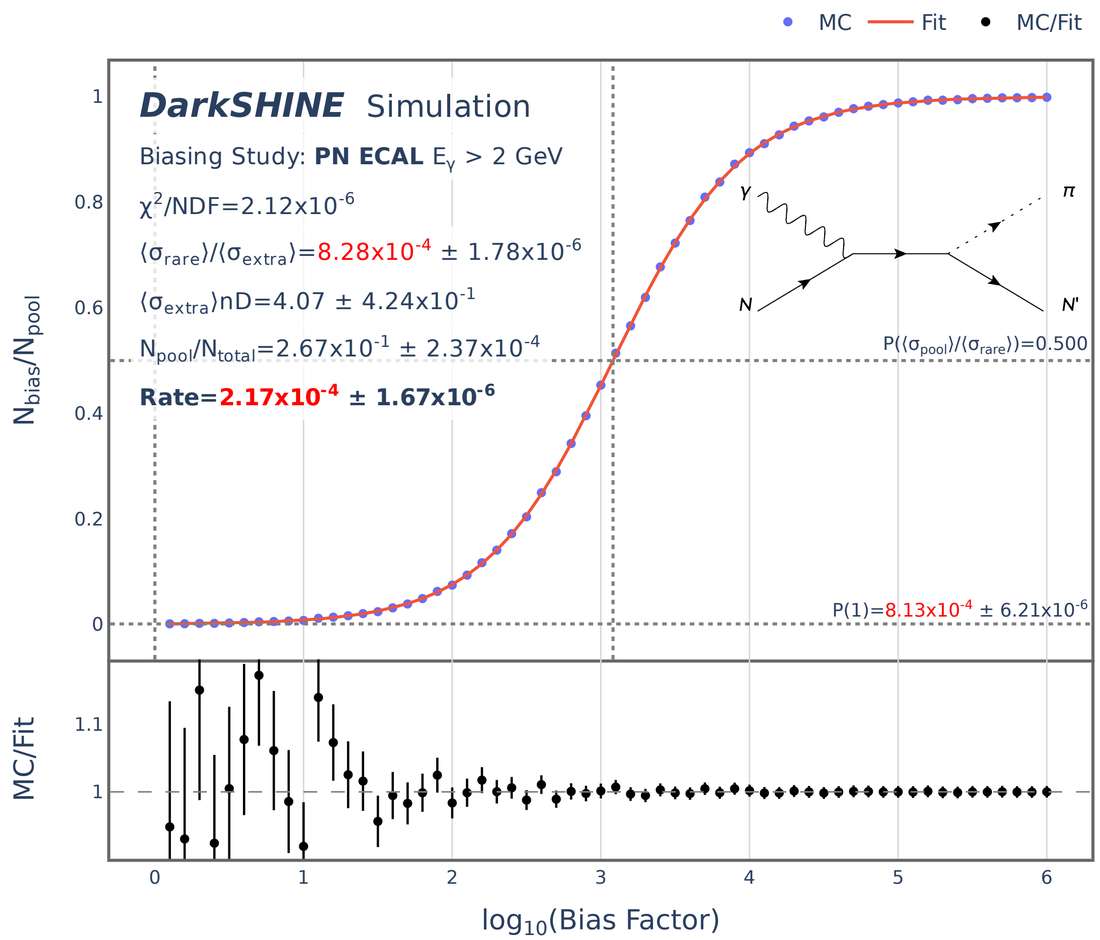


ECAL γ process

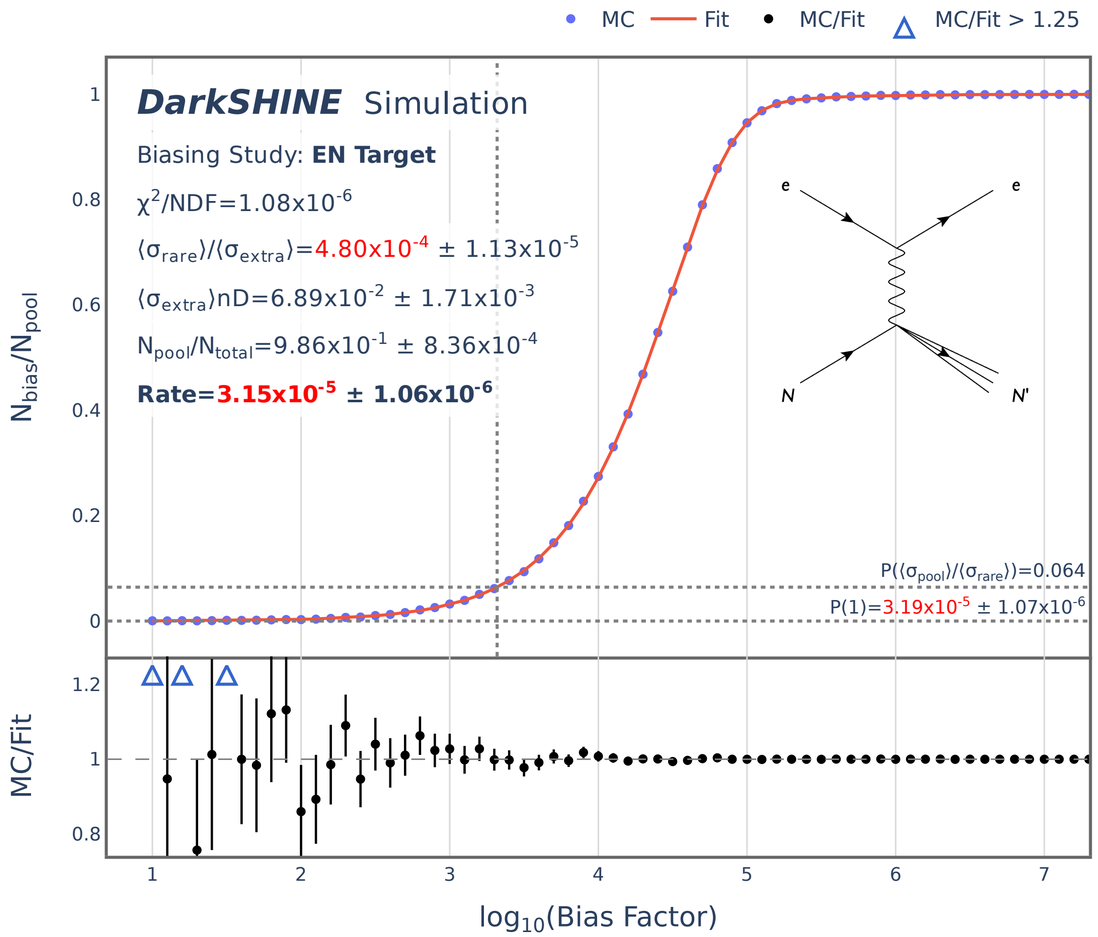
$$P(x) = Br(x) \times \left(1 - e^{-(x\langle\sigma_r\rangle + \langle\sigma_e\rangle)nD}\right)$$

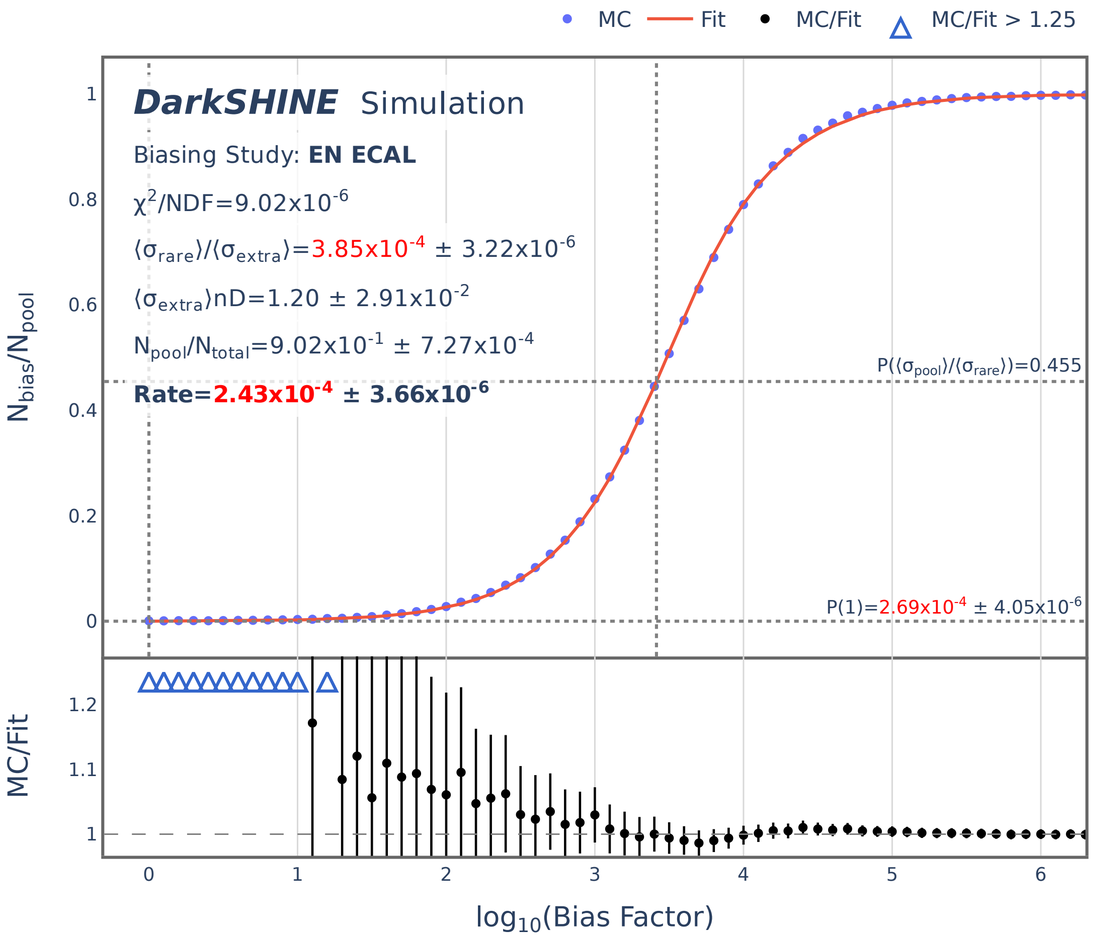
$\langle\sigma\rangle nD \gg Br$: P(1) ~ Br





Electron Nuclear Target / ECAL

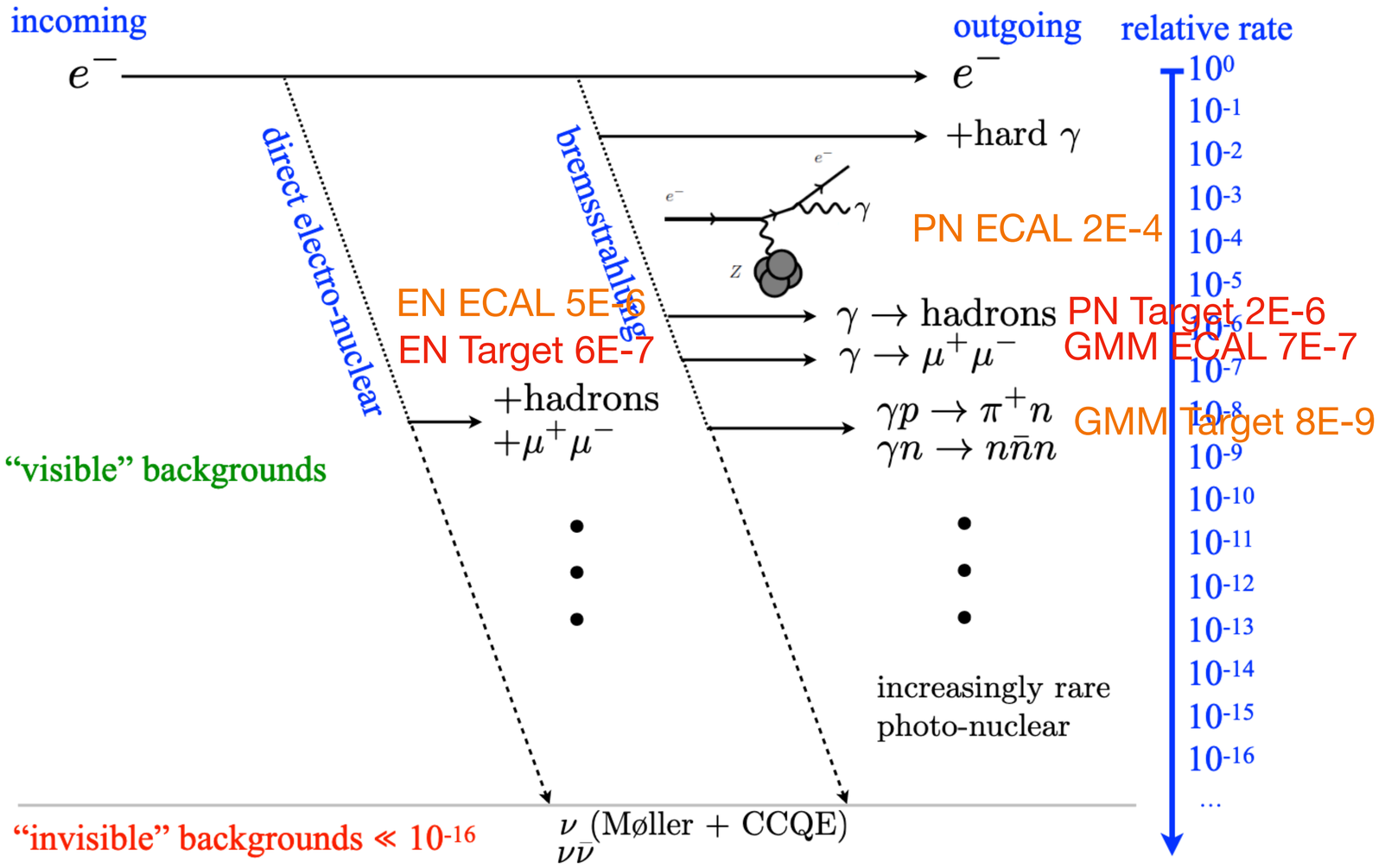




Rate

	Rate	Rate Error	**Filter (ΔE>2GeV) Efficiency**	Filter Efficency Error	**Rate × ε_{filter}**	Inclusive Rate (~1e7)	Ratio of Inclusive / Fit
GMM Target (E_γ>2GeV)	**8.1302E-09**	9.1720E-11	**1.0**	0.0	**8.1302E-09**	*0*	*0.0*
GMM ECAL (E_γ>2GeV)	**6.9125E-07**	1.4577E-09	**1.0**	0.0	**6.9125E-07**	*1.04E-06*	*1.5*
PN Target (E_γ>2GeV)	**1.7332E-06**	1.7523E-08	**1.0**	0.0	**1.7332E-06**	*4.65E-06*	*2.7*
PN ECAL (E_γ>2GeV)	**2.1723E-04**	1.6695E-06	**1.0**	0.0	**2.1723E-04**	*2.87E-04*	*1.3*
EN Target (E_e>2GeV)	**3.1459E-05**	1.0566E-06	**1.8729E-02**	3.1110E-05	**5.8921E-07**	*1.4600E-06*	*2.5*
EN ECAL (E_e>2GeV)	**2.4271E-04**	3.6583E-06	**1.8899E-02**	9.4327E-05	**4.5871E-06**	*3.3300E-06*	*0.7*

Comparison with LDMX



$N_{record} = N_{BeamOn} \times P_{pool} \times \varepsilon_{filt+trig} = N_{EOT} \times Rate \times \varepsilon_{filt+trig}$

Production *w/ truth trigger*

Truth Trigger: $P_{miss} > 1$ GeV, $E_{ECAL} < 3$ GeV

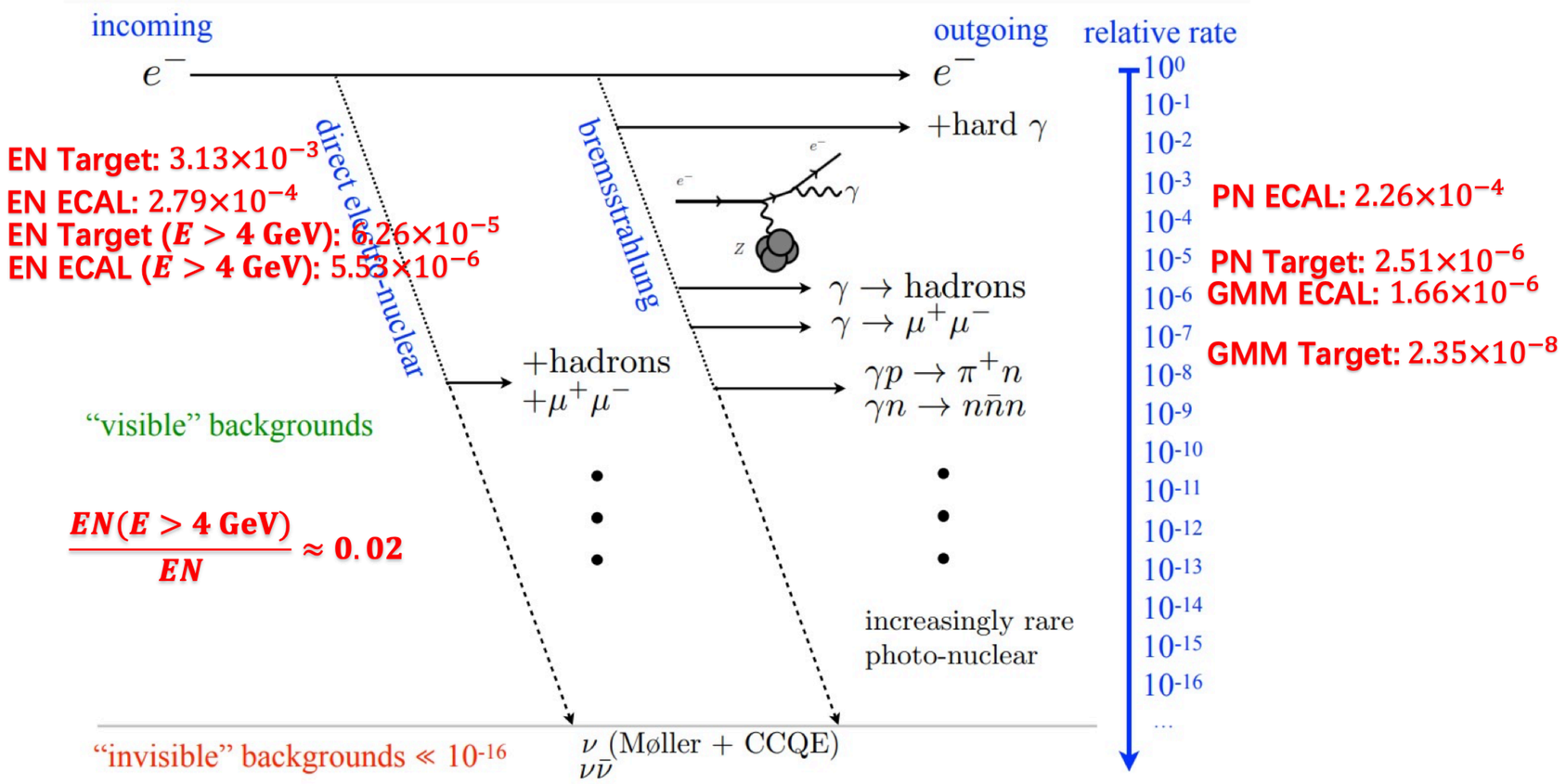
	EOT	Rate	P_{pool}	Beam On	Time / 16000 cpu [hour]	Total Cost *(¥0.025 / cpu hour)*	Total Disk [TB]	$\varepsilon_{filt+trig}$	Time / Event [ms]	Disk / Event [KB]
GMM Target (E_γ>2GeV)	**1E+15**	8.13E-09	6.85%	**1.1860E+08**	**0.01**	**5.33**	**0.25**	99.97%	6.48	33.40
GMM ECAL (E_γ>2GeV)	**1E+15**	6.91E-07	26.67%	**2.5919E+09**	**1.27**	**507.02**	**10.20**	52.14%	28.17	30.40
PN Target (E_γ>2GeV)	**1E+15**	1.73E-06	6.85%	**2.5289E+10**	**5.08**	**2031.97**	**98.86**	84.78%	11.57	72.24
EN Target (ΔE>2GeV)	**1E+15**	3.15E-05	98.57%	**3.1915E+10**	**3.67**	**27.49**	**9.12**	0.43%	6.62	71.79
EN ECAL (ΔE>2GeV)	**1E+15**	2.43E-04	90.15%	**2.6922E+11**	**40.75**	**308.08**	**298.54**	1.89%	8.72	70.06
Total					**50.78**	**2879.89**	**416.97**	DAna: ~14TB		

¥0.55 / TB / day

Backup

2021 ver

Comparison with LDMX

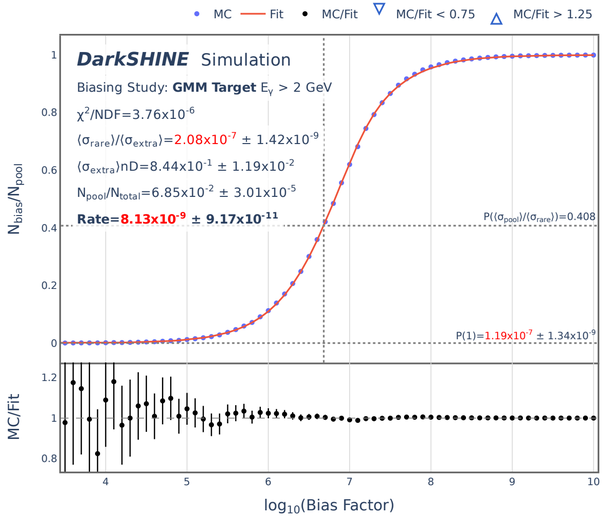


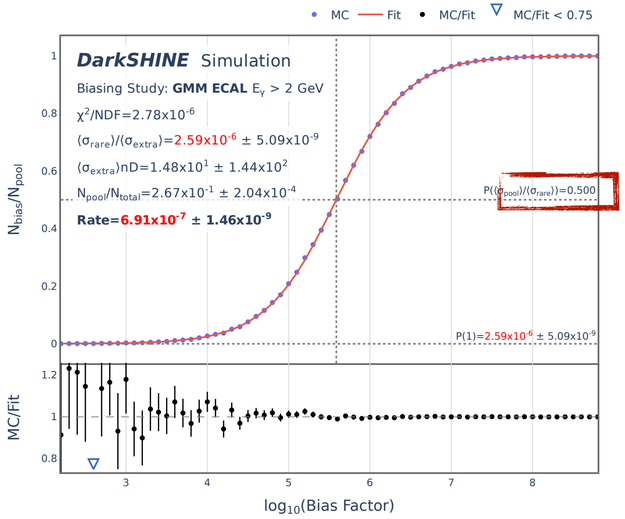
EN Target: 3.13×10^{-3}
EN ECAL: 2.79×10^{-4}
EN Target ($E > 4$ GeV): 6.26×10^{-5}
EN ECAL ($E > 4$ GeV): 5.53×10^{-6}

$$\frac{EN(E > 4\ \text{GeV})}{EN} \approx 0.02$$

PN ECAL: 2.26×10^{-4}
PN Target: 2.51×10^{-6}
GMM ECAL: 1.66×10^{-6}
GMM Target: 2.35×10^{-8}

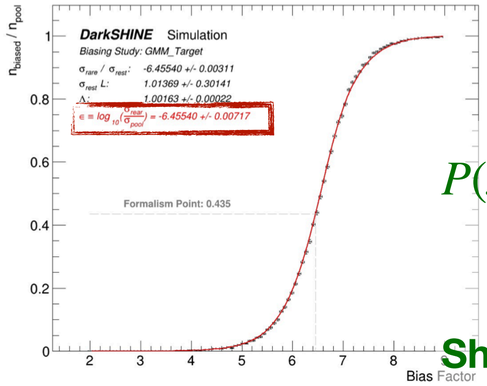
New rate calculation

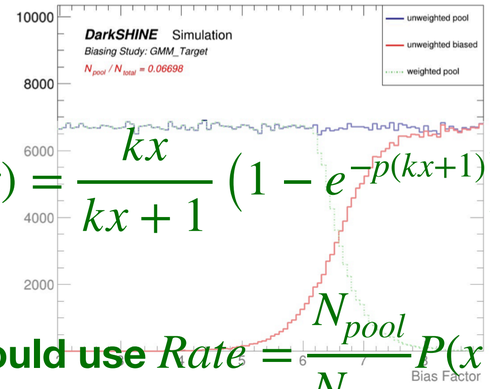




- σ_{biased} : cross-section of biased events
- σ_{pool} : cross-section of interested events
- ϵ : analog ratio between biased and interested eve
- Formalism point : ratio corresponded to the ϵ

GMM Target





$$P(x) = \frac{kx}{kx+1}\left(1 - e^{-p(kx+1)}\right), \quad k \equiv \frac{\sigma_{rare}}{\sigma_{extra}}, p \equiv \sigma_{extra} D$$

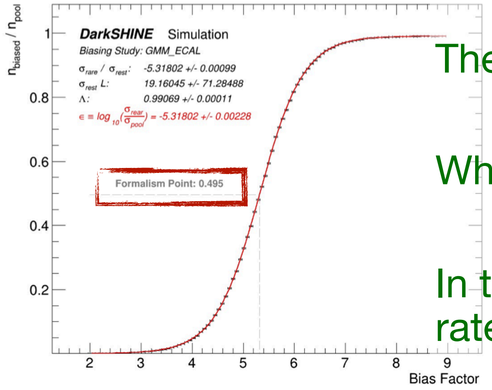
Should use $Rate = \frac{N_{pool}}{N_{total}} P(x=1)$ **instead of** $Rate = \frac{N_{pool}}{N_{total}} \frac{\sigma_{rare}}{\sigma_{pool}}$

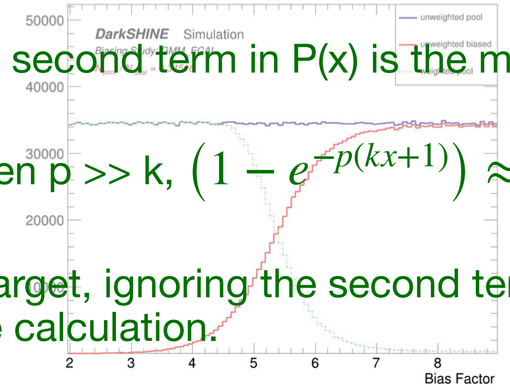
rel. to inclusive: 2.35×10^{-8}

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GMM ECAL

The first term in P(x) is Branching ratio $\frac{\sigma_{rare}}{\sigma_{pool}} = \frac{\sigma_{rare}}{\sigma_{rare} + \sigma_{extra}}$





The second term in P(x) is the material length effect.

When p >> k, $\left(1 - e^{-p(kx+1)}\right) \approx 1$, i.e. ECAL, can ignore second term.

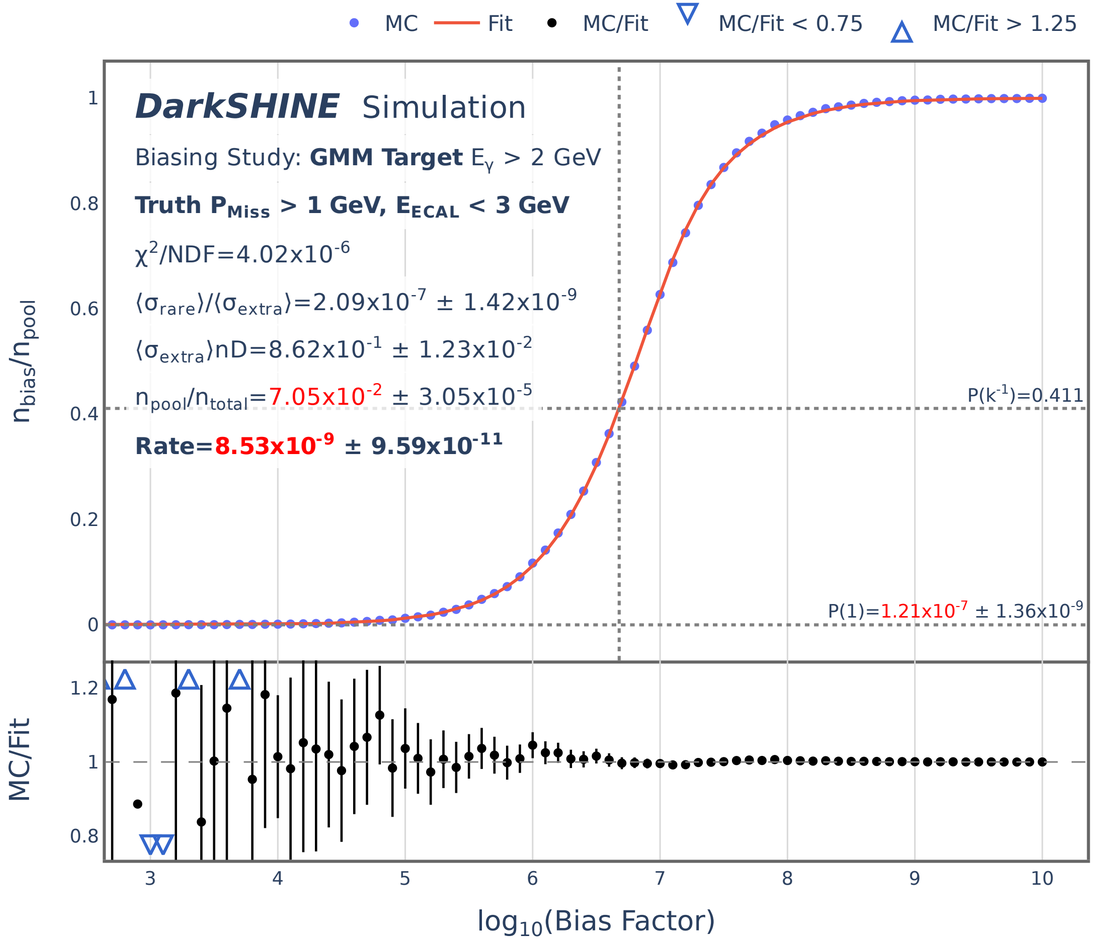
In target, ignoring the second term will cause a factor of **~2** to **~15** in rate calculation.

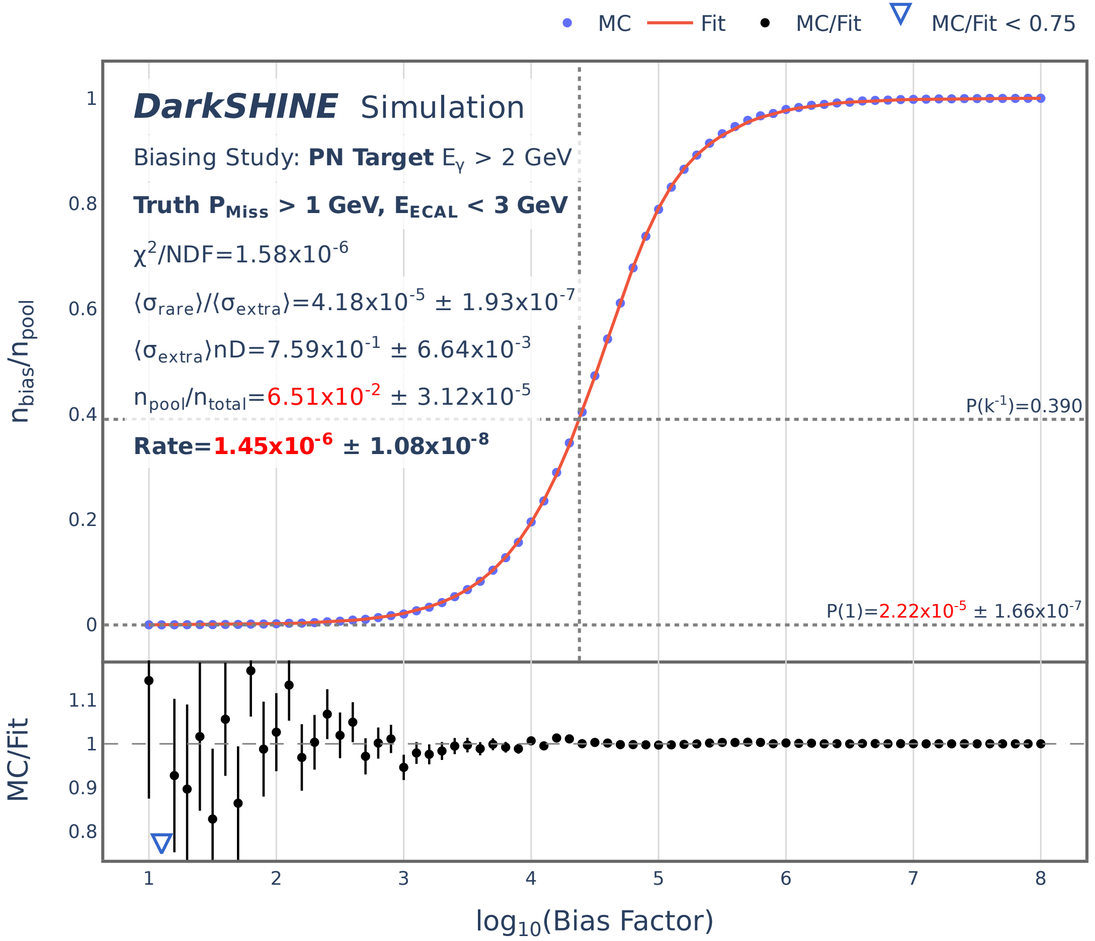
rel. to inclusive: 1.66×10^{-6}

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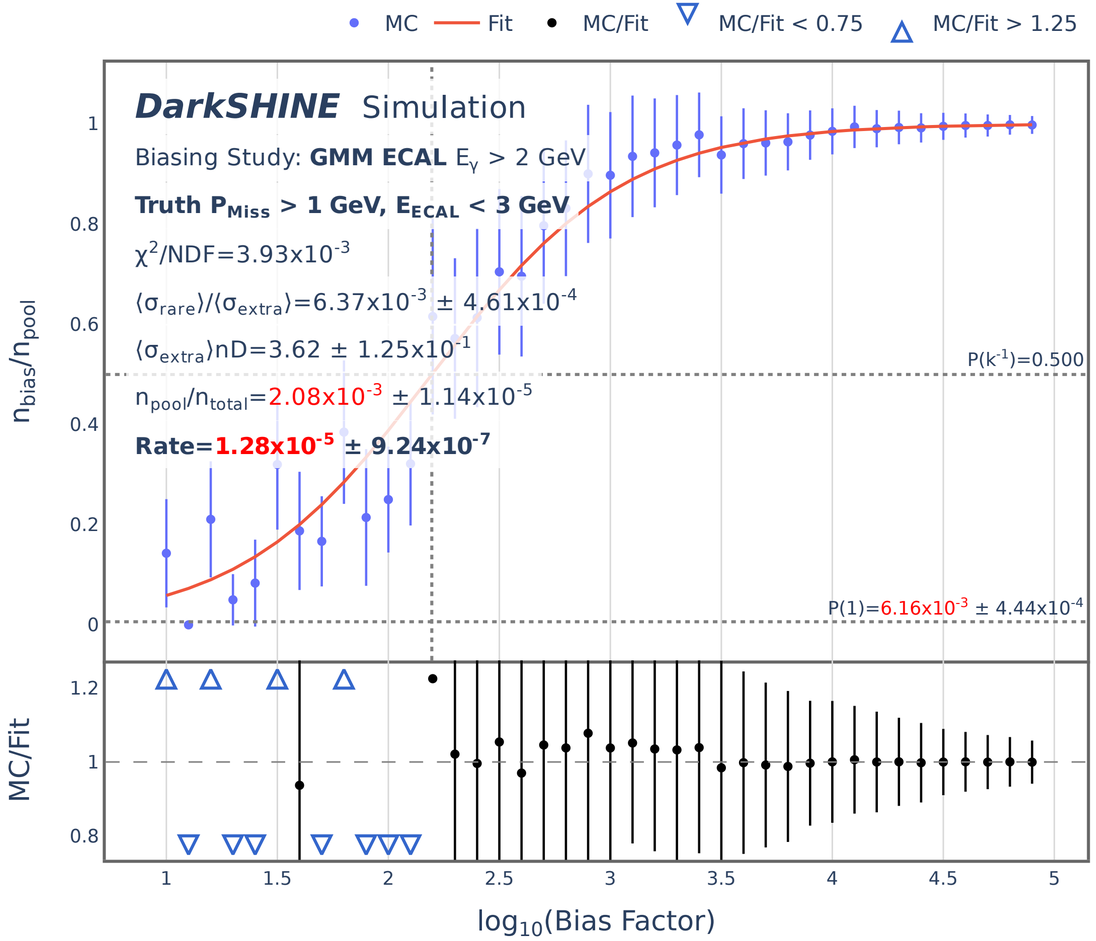
Target γ process w/ Truth Trigger

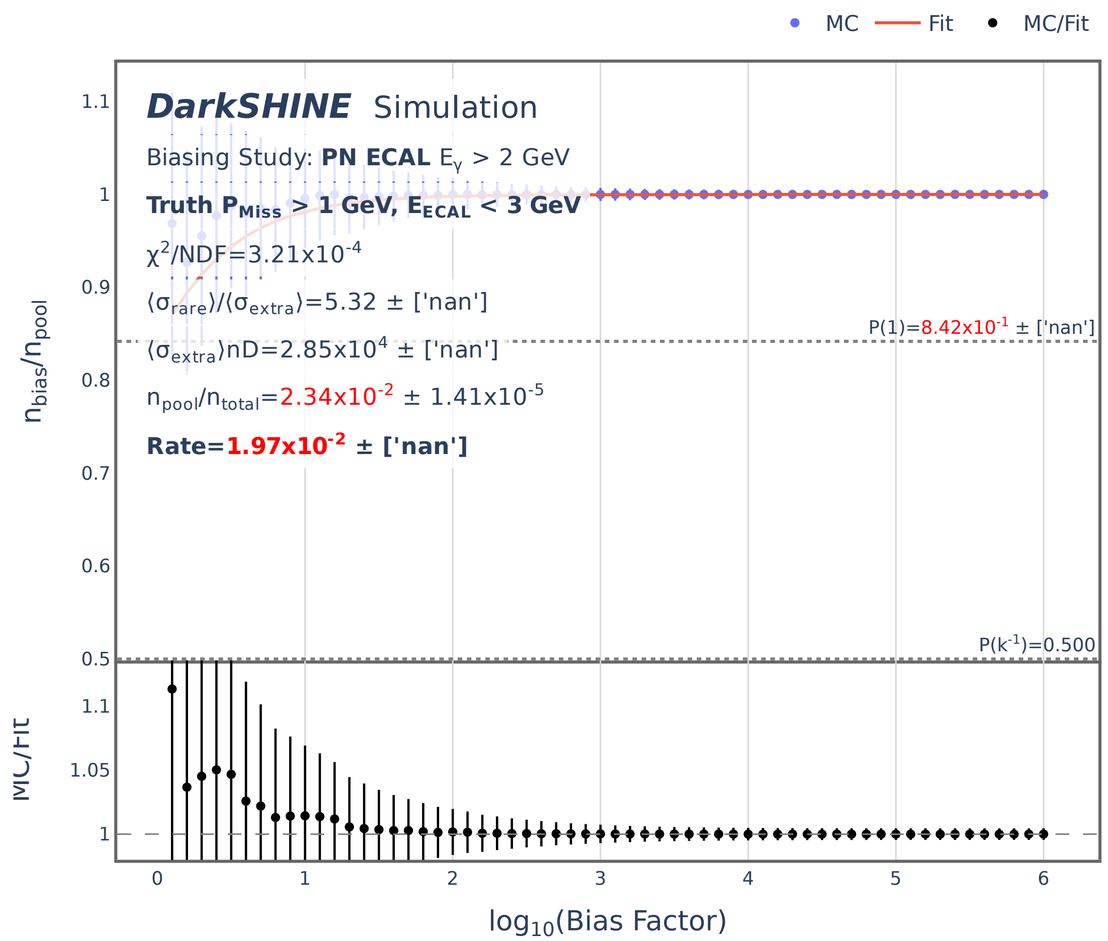
- Following fit results have **no** physical meaning





ECAL γ process w/ Truth Trigger





Target / ECAL Electron Nuclear w/ Truth Trigger

